



AIMS

African Institute for
Mathematical Sciences
RWANDA

Introduction to Probability and Statistics

Lecture 9: Properties of Estimators

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1 General properties of estimators

2 Convergence

Bias of an estimator

Notation: $g_n(X_1, \dots, X_n) = g_n(\underline{X})$

Definition

An estimator $g_n(\underline{X})$ of θ is an **unbiased estimator** if

$$E[g_n(X_1, \dots, X_n)] = \theta, \text{ for all } \theta \in \Theta.$$

$b_n(\theta) = E[g_n(\underline{X})] - \theta$ is called the **bias** of the estimator $g_n(\underline{X})$.

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Exercise

- Let $X_i \sim \mathcal{B}(N, p)$, $i = 1, \dots, n$.
Is $\hat{p} = \bar{X}_n/N$, the MLE, an unbiased estimator?

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Exercise

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Is $\hat{p} = \bar{X}_n/N$, the MLE, an unbiased estimator?
- Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$, $i = 1, \dots, n$.
Is $s_X^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$ an unbiased estimator of σ^2 ?

Bias of an estimator

Definition

We say that $g_n(\underline{X})$ is an **asymptotically unbiased estimator** of θ if

$$\lim_{n \rightarrow +\infty} b_n(\theta) = 0, \text{ for all } \theta \in \Theta.$$

Exercise:

Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$, $i = 1, \dots, n$.

- Show that s_X^2 is an asymptotically unbiased estimator of σ^2 .

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- Show that s_X^2 is an asymptotically unbiased estimator of σ^2 .
- Build an unbiased estimator of σ^2 .

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Exercise:

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- Show that s_X^2 is an asymptotically unbiased estimator of σ^2 .
- Build an unbiased estimator of σ^2 .

$$\hat{\sigma}_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \text{ is an unbiased estimator of } \sigma^2.$$

Convergence of a sequence of r.v.'s

Let Z_n be a sequence of r.v. $\rightarrow g_n(\underline{X})$ is a sequence of r.v.

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Convergence in probability

Z_n converges to Z **in probability** iff,

$$\forall \varepsilon > 0, \lim_{n \rightarrow +\infty} P(|Z_n - Z| > \varepsilon) = 0.$$

Notation: $Z_n \xrightarrow{P} Z$

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Convergence in quadratic mean or in mean square

Z_n converges to Z in quadratic mean ($Z_n \xrightarrow{q.m.} Z$) iff,

$$\forall \varepsilon > 0, \lim_{n \rightarrow +\infty} E(Z_n - Z)^2 = 0.$$

Notation: $Z_n \xrightarrow{q.m.} Z$

Convergence in Probability

Definition

Let $X_1, \dots, X_n$ be n r.v.'s with a distribution P_θ , $\theta \in \Theta$, where θ is unknown belonging to Θ , then the sequence $g_1, g_2, \dots$ is called a **consistent** sequence of estimators of θ if

$$g_n(\underline{X}) \text{ converges in probability to } \theta \quad (g_n(\underline{X}) \xrightarrow{P} \theta)$$

that is:

$$\forall \varepsilon > 0, \lim_{n \rightarrow +\infty} P(|g_n(\underline{X}) - \theta| > \varepsilon) = 0, \quad \forall \theta \in \Theta.$$

$g_n(\underline{X})$ is said to be a **consistent estimator** of θ .

Convergence in Probability

Example:

Let $X_1, \dots, X_n$, be n independent r.v. such that $X_i \sim \mathcal{N}(\mu, \sigma^2)$, then $\bar{X}_n$ converge in probability to μ .

That is to say:

$$\forall \varepsilon > 0, \lim_{n \rightarrow +\infty} P(|\bar{X}_n - \mu| > \varepsilon) = 0.$$

Proof: $\bar{X}_n \sim \mathcal{N}(\mu, \sigma^2/n)$ and $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = Z_n \sim \mathcal{N}(0, 1)$.

Then

$$\begin{aligned} P(|\bar{X}_n - \mu| > \varepsilon) &= P(|Z_n| > \frac{\varepsilon}{\sigma} \sqrt{n}) = 1 - P(|Z_n| \leq \frac{\varepsilon}{\sigma} \sqrt{n}) \\ &= 2 - 2\Phi\left(\frac{\varepsilon}{\sigma} \sqrt{n}\right). \end{aligned}$$

But $\lim_{n \rightarrow +\infty} \Phi\left(\frac{\varepsilon}{\sigma} \sqrt{n}\right) = 1$; that shows the result.

Two useful inequalities:

Markov's Inequality

Let X be a positive r.v. such that $EX < +\infty$. Then:

$$\forall \delta > 0, P(X > \delta) \leq \frac{EX}{\delta}.$$

Chebyshev's Inequality

Let X be a r.v. such that $EX < +\infty$ and $Var(X) < +\infty$. Then:

$$\forall \varepsilon > 0, P(|X - EX| > \varepsilon) \leq \frac{Var(X)}{\varepsilon^2}.$$

Proof: Apply the Markov's Inequality to $Y = (X - EX)^2$.

Convergence in Probability

Exemple (cont.):

Let $X_1, \dots, X_n$, be n independent r.v. such that $X_i \sim \mathcal{N}(\mu, \sigma^2)$, then $\bar{X}_n$ converge in probability to μ .

Applying Chebytshev's inequality to $\bar{X}_n$:

$$P(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\sigma^2}{n\varepsilon} \longrightarrow 0. \quad \rightarrow \bar{X}_n \xrightarrow{P} \mu.$$

Exercise

Let $X_1, \dots, X_n$, be n independent r.v. such that $X_i \sim \mathcal{P}(\lambda)$. Show that $\bar{X}_n$ converge in probability to λ .

Consistency in quadratic mean

Definition

Let $X_1, \dots, X_n$ be n r.v.'s with a distribution P_θ , $\theta \in \Theta$.

Let $g_n(\underline{X})$ be a statistic of $\underline{X}$.

We say that $g_n(\underline{X})$ is **consistent in quadratic mean** ($g_n(\underline{X}) \xrightarrow{q.m.} \theta$) if

$$\lim_{n \rightarrow +\infty} E[(g_n(\underline{X}) - \theta)^2] = 0, \quad \forall \theta \in \Theta.$$

The quantity $E[(g_n(\underline{X}) - \theta)^2]$ is called the **mean squared error (MSE)** of the estimator.

Consistency in quadratic mean

Example:

Let $X_1, \dots, X_n$, be n independent r.v. such that $X_i \sim \mathcal{N}(\mu, \sigma^2)$, then $\bar{X}_n$ converge in quadratic mean to μ .

Proof:

$$E(\bar{X}_n - \mu)^2 = \text{Var}(\bar{X}_n) = \sigma^2/n \longrightarrow 0. \quad \Rightarrow \quad \bar{X}_n \xrightarrow{q.m.} \mu.$$

Exercise

Let $X_1, \dots, X_n$, be n independent r.v. such that $X_i \sim \mathcal{P}(\lambda)$. Show that $\bar{X}_n$ converge in quadratic mean to λ .

Consistency in quadratic mean

Theorem

$g_n(\underline{X})$ is **consistent in quadratic mean** iff

- (i) $\lim_{n \rightarrow +\infty} \text{Var}[g_n(\underline{X})] = 0$
- (ii) $g_n(\underline{X})$ is asymptotically unbiased.

Proof:

Lemma

$$E[(g_n(\underline{X}) - \theta)^2] = \text{Var}[g_n(\underline{X})] + [b_n(\theta)]^2, \quad \forall \theta \in \Theta.$$

➡ *Bias-Variance Trade-Off*

Consistency in quadratic mean

Theorem

$g_n(\underline{X})$ is consistent **in quadratic mean**, $g_n(\underline{X})$ converges **in probability**.

$$\left(g_n(\underline{X}) \xrightarrow{q.m.} \theta\right) \implies \left(g_n(\underline{X}) \xrightarrow{P} \theta\right).$$

Proof:

Apply Markov's inequality to $|g_n(\underline{X}) - \theta|^2$.